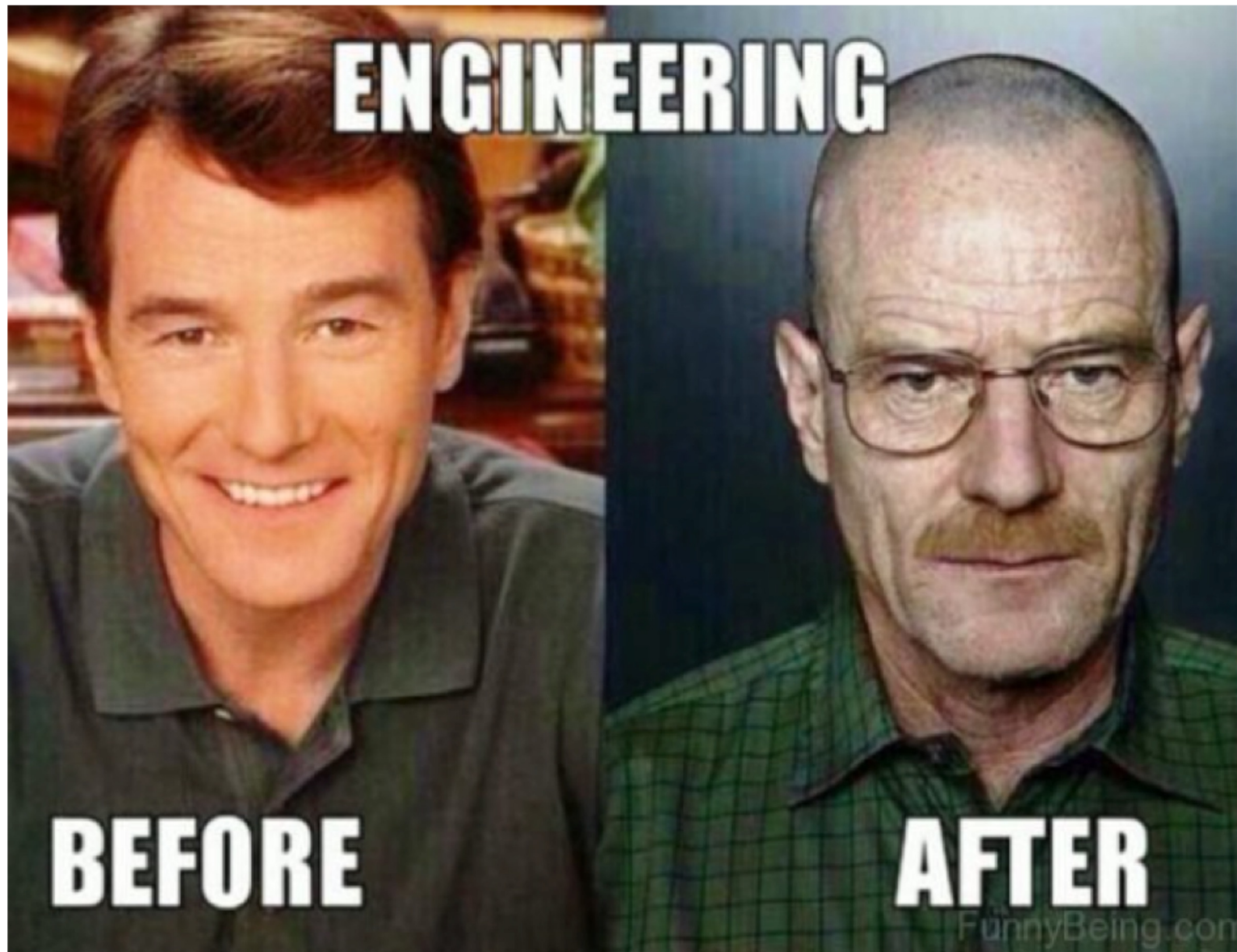




INTRODUCTION

Com Eng Math II



Warning: 3rd year CP is no joke. Plan your life accordingly!

Communications

Mycourseville

password: 2026hamham

My Courses Online Courses Evaluation Center Join a course Back Office

Registering / Enrolling

As a Student to a Course As a new Instructor

Search for a course to enroll as a student

Course Number: 2110203

Year/Semester: 2026/1

Q Search for Courses

Course search result

2110203.01 2026/1 was found.

1 course found

Computer Engineering Mathematics II [Section 1]

To register as Punnarai Siricharoen

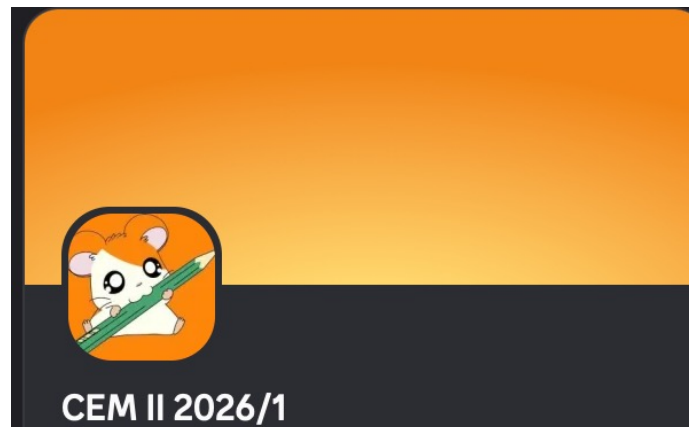
Enter Course Password

2026hamham

Request to register

Discord

<https://discord.gg/H7rMBJ5jA>



Format

Hybrid (on-site and online + recordings)

211203 Com Eng Math 2

Homework & Assignment

Midterm + Quiz + Final

Participation and in class activities

Late turn-ins and evaluation

การส่งงานสาย

สายไม่เกิน 6 ชม. -10% ของคะแนนเต็มของงาน

สายไม่เกิน 24 ชม. -20% ของคะแนนเต็มของงาน

ถ้าส่งสายเกิน 24 ชม. จะไม่ได้รับการตรวจ

เกณฑ์การวัดผล

Com Eng Math 2

Midterm + Quiz + Final = 30 + 25 + 30 = 85%

Homework 10% (13x1 3% extra)

Participation and in class activities 5% (0.5x10 + bonus)

Quiz and Exams

2110203 CEM II

- การสอบทั้งหมด in-person

Homework

- Only a couple of problems in the homework will be graded (declared in the homework)
- The rest should be used for practice
- Solutions will be provided after 24 hours of the deadline
 - Out: Monday
 - Due: Sunday night
- Submission
 - Code: Ipython Notebook
 - Writeup: **one single PDF only**. Can write or type, but final submission must be in PDF format. (No zip, jpg, png, etc.)

Getting help

- Discord
 - Text Channels
 - Can use Voltaire for anonymous questions
 - **Direct PMs will be ignored** if not a personal question
- In person
 - Tuesday 4-5pm, Sky Cafe
- **No help** will be offered on Sundays
 - Do your homework early!

Academic integrity

- You shall **not copy** others.
 - For homeworks, violation will result in **0** for both parties.
 - For exam/quizzes, violations result in an **F** and other penalties according to Faculty of Engineering guidelines.
- You can collaborate or discuss but:
 - You must declare the person/Chat-GPT/Claude/Gemini you collaborate with in the homework
 - You shall not show exact write-up with each other
 - Can discuss ideas but not implementation
 - Can compare final answers

นโยบายการใช้ AI และการลอกผลงาน

- งานทุกงานของนิสิตควรตอบด้วยคำพูดของตนเอง ห้ามการลอกผลงานจากเพื่อนหรือ AI โดยไม่มีการไต่ตรอง นิสิตสามารถพูดคุยกับเพื่อนหรือ AI เพื่อให้การช่วยเหลือในการทำการบ้านได้ (ยกเว้นจะมีการแจ้งพิเศษเฉพาะการบ้าน) นิสิตต้องมีความรับผิดชอบต่อคำตอบของตนเอง เช็ความถูกต้องของ AI ต้องตอบ โดยคำพูดของตนเอง **ห้าม** ตอบในลักษณะของการคัดลอกคำตอบต่อคำถามโดยเด็ดขาด ถ้าพบว่ามี การคัดลอกคำตอบโดยตรง จะปรับเป็นศูนย์คะแนนในการบ้านนั้น

Signal
Processing

สัปดาห์ที่	เนื้อหา	การบ้านและแลป
1 - 3/8	Signals & Transformations	Assignment 1
2 - 10/8	Convolution	Assignment 2 (Convolution)
3 - 17/8	Continuous-time fourier transform	Assignment 3 (FT)
4 - 24/8	Discrete-time Fourier transform (DTFT)	Assignment 4 (DTFT)
5 - 31/8	Discrete Fourier Transform (DFT)	Assignment 5 (DFT)
6 - 7/9	Applications: Sampling, Filtering, Image Processing, etc	Assignment 6 (Applications)

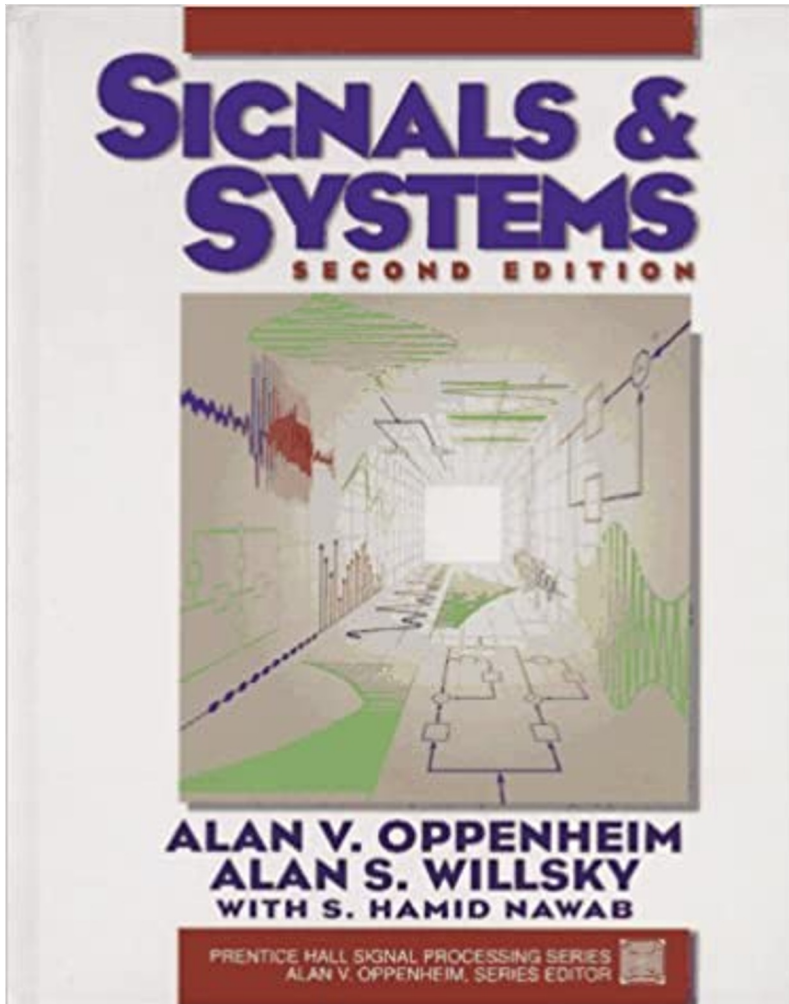
Optimization

7 - 14/9	Linear Programming & Simplex method	Assignment 7 (Optimization)
8 - 21/9	Midterm week - Midterm covers up to Week 6	
9 - 28/9	Simplex method II	Assignment 8 (Simplex)
10 - 5/10	Integer optimization	Assignment 9 (Integer prog)

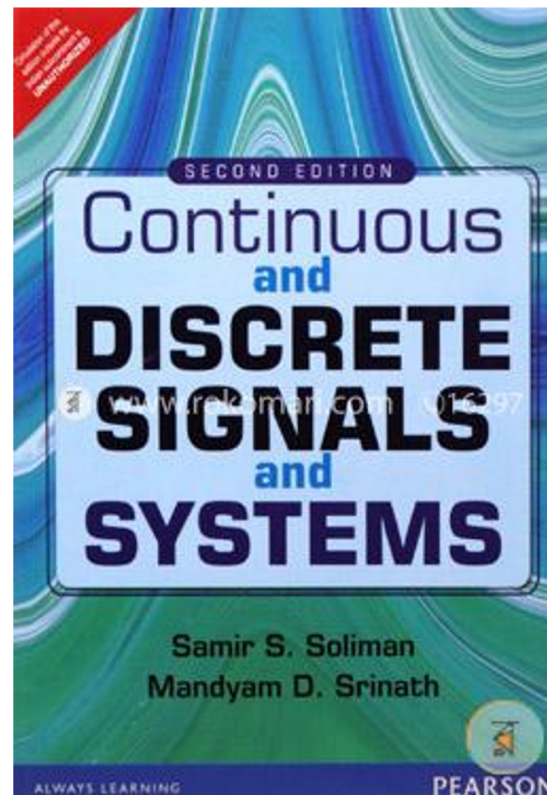
Statistics

11 - 12/10	Probability recap	
12 - 19/10	Optimization Quiz - covers week 7 to 10	
13 - 26/10	CLT, LLN, Sampling, Estimation	Assignment 10 (Prob & simulation)
14 - 2/11	Testing	Assignment 11 (Testing)
15 - 9/11	A/B testing	Assignment 12 (A/B)
16 - 16/11	Multi-arm bandits	Assignment 13 (multi-arm)
	Final exam - cover week 11 to 15	

Books

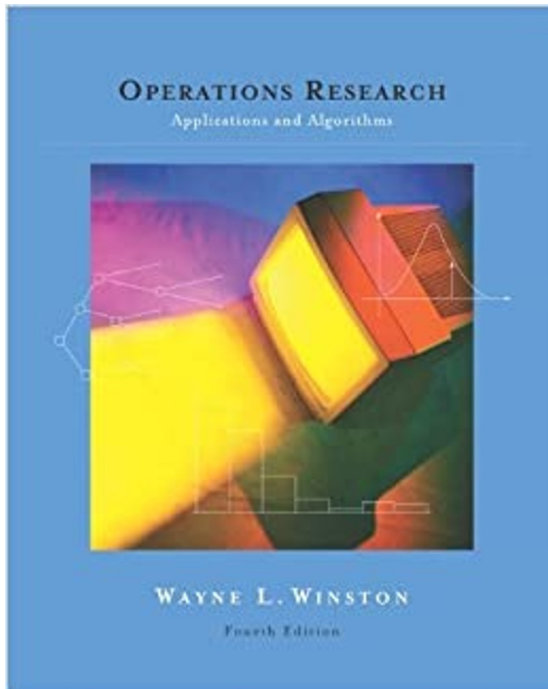


Alan V. Oppenheim, Alan S. Willsky, Ian T. Young. Signals and Systems



Samir S. Soliman and Mandyam D. Srinath. Continuous and discrete signals and systems

Books



Wayne L. Winston. Operations research

Books

MITOPENCOURSEWARE MASSACHUSETTS INSTITUTE OF TECHNOLOGY				
Subscribe to the OCW Newsletter f i t v				
Home Contact Us				
FIND COURSES For Educators Give Now About Search Search Tips				
Home » Courses » Mathematics » Introduction to Probability and Statistics » Readings				
Readings				
Listed in the following table are assigned readings and reading questions that students were expected to complete prior to attending class sessions. The reading questions are multiple choice or numerical answer questions. Students received instant feedback and could make multiple attempts. Another form of interactive assessment used in the course, Problem Set Checker, can be viewed on the Assignments page.				
COURSE HOME	WEEK #	SES #	READINGS	INTERACTIVE READING QUESTIONS
SYLLABUS	Probability			
CALENDAR				
INSTRUCTOR INSIGHTS				
READINGS	1	C1	1a: Introduction (PDF) 1b: Counting and Sets (PDF)	Reading Questions for 1a Reading Questions for 1b
CLASS SLIDES		C2	2: Probability: Terminology and Examples (PDF) R Tutorial 1A: Basics R Tutorial 1B: Random Numbers	Reading Questions for 2
ASSIGNMENTS		C3	3: Conditional Probability, Independence and Bayes' Theorem (PDF)	Reading Questions for 3
EXAMS	2	C4	4a: Discrete Random Variables (PDF) 4b: Discrete Random Variables: Expected Value (PDF)	Reading Questions for 4a Reading Questions for 4b

<https://ocw.mit.edu/courses/mathematics/18-05-introduction-to-probability-and-statistics-spring-2014/readings/>



Ron Kohavi, Diane Tang, Ya Xu. Trustworthy Online Controlled Experiments
 John Myles White. Bandit Algorithms for Website Optimization

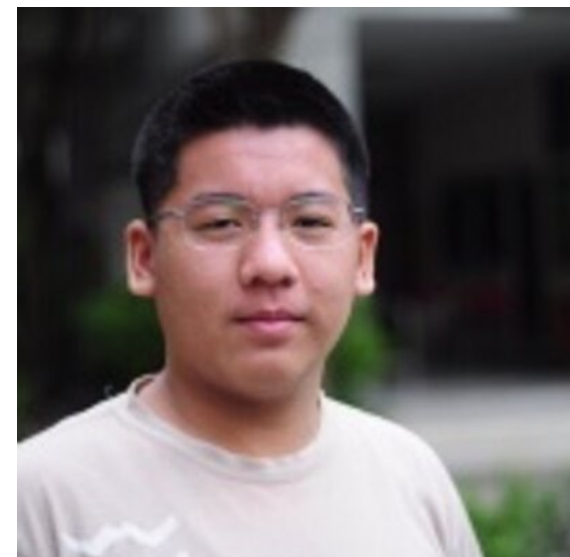
Instructors



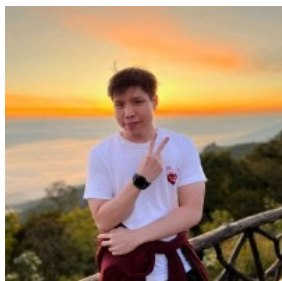
ผศ.ดร.พรพรรณราย ศิริเจริญ



ผศ.ดร. เอกพล ช่างสูวนิช



อ. ดร. ณัท ภาวสันต์



พีศิริ



พีตะ



พีหยก



พีหลิว



พีกีตาร์



พีตัน



พีหลิว



Com Eng Math II

Course philosophy

Math with applications in computer engineering

Theory + practice

Various aspects of

“What’s the best way to _____?”



Signal processing

How to analyze and manipulate data (sounds, images, stock prices) in another basis?

Examples

- Sound processing

- Image processing

- Control systems (robots, motors)

Optimization

How to allocate resources?

How to optimize a certain process?

Examples

Task scheduling

Resource allocation

Where to place a store

Machine learning

Linear programming as a framework for automating decisions

Previously, feature approval was a manual process where teams would spend valuable time calculating how many features we could support and analyzing what the return on investment was for increasing the capacity of our services. In a company like Facebook — where we have multiple models being continuously iterated on — this approach does not scale. By framing our services as a system of linear equations, we take a complex interconnected system and simplify it into basic relationships that are easily communicated. By doing this, we can make smarter decisions about the features we deploy and the infrastructure we invest in.

<https://engineering.fb.com/2021/07/29/data-infrastructure/linear-programming/>

Statistics (testing)

How to be certain (statistically) that a certain process is better?

How to learn about user behavior in a process?

Example

User behavior optimization in a website

Is model A better than model B?

Vaccine efficacy

Drug effectiveness

ถอนแล้ว! งานวิจัยฟ้าทะลายโจรลดปอดบวมโควิดได้ หลักฐานยังไม่พอ

<https://today.line.me/th/v2/article/pBRO1q>



สยามรัฐ

อัปเดต 1 วันที่แล้ว • เผยแพร่ 1 วันที่แล้ว • สยามรัฐออนไลน์

ติดตาม

